Abhinay Goud Adhire

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EDUCATION

Master of Science in Computer Science — California State University, San Bernardino *May 2026*

Relevant Coursework: Big Data Management, Database Systems, Artificial Intelligence, Numerical Computation

Bachelor of Technology in Computer Science and Engineering — GITAM University, Hyderabad, India *May 2023*

SUMMARY

Recent Computer Science graduate (M.S., May 2026) with hands-on experience building orchestrated data pipelines with Apache Spark, Airflow, and dbt. Comfortable across the full ELT lifecycle — from raw ingestion and workflow orchestration through medallion-layered transformations to analytics-ready marts — with an emphasis on data quality, idempotency, and pipeline portability.

TECHNICAL SKILLS

Languages: Python, SQL

Big Data Processing: Apache Spark (DataFrame API & Spark SQL), Spark Structured Streaming

Orchestration & Transformation: Apache Airflow, dbt (staging → marts, dbt tests)

Lakehouse & Storage: Delta Lake, medallion architecture (Bronze/Silver/Gold), Databricks, PostgreSQL

Data Modeling: ELT pipeline design, denormalized fact tables, window functions

Tooling: Jupyter/Databricks notebooks, Docker, Git/GitHub

Foundations: Data structures & algorithms, distributed systems, relational database design, numerical computation

PROJECTS

E-commerce Sales Analytics Pipeline *Big Data Management Course*

Apache Spark, Delta Lake, Databricks, PySpark, Spark SQL, Spark Structured Streaming

GitHub: github.com/abhinay2001/ecommerce-sales-pipeline

- Designed and built an end-to-end ELT pipeline on Apache Spark and Delta Lake, ingesting and transforming 1.55M+ records across 9 relational source tables from a real-world e-commerce dataset.
- Implemented a medallion lakehouse (Bronze/Silver/Gold) with Delta Lake ACID transactions and schema enforcement, joining 5 source tables into a partitioned, denormalized fact table for analytics use.
- Wrote transformations in both the Spark DataFrame API and Spark SQL, including multi-table joins and window functions (running sums, LAG, ntile) to power monthly revenue trends and RFM customer segmentation.
- Built a Spark Structured Streaming job computing 5-minute tumbling-window revenue aggregations on a simulated live-order feed, using a connector interchangeable with Kafka in production.
- Delivered gold-layer analytics — RFM segmentation, cohort retention, and market-basket affinity — surfacing findings such as ~3% month-1 retention across R$15.7M in analyzed revenue.

Weather Forecast Accuracy Tracker *Independent Project*

Apache Airflow, dbt, PostgreSQL, Python, Docker

GitHub: github.com/abhinay2001/weather-forecast-accuracy

- Built an Airflow-orchestrated pipeline ingesting daily weather forecasts and historical actuals for 12 cities from the Open-Meteo API into Postgres, with idempotent upserts guarding against duplicate DAG reruns.
- Modeled the warehouse in dbt across staging, intermediate, and mart layers, joining forecast and actual observations by lead time and validating the transformations with dbt tests.
- Designed a trailing-window actuals sweep to handle the ERA5 reanalysis data's processing lag, and a historical-forecast backfill script (Open-Meteo's Previous Runs API) to seed 90 days of real overlapping forecast/actual data immediately.
- Analyzed 8,000+ forecast/actual pairs across 7 lead times, finding average forecast error roughly doubles from 1 day out (1.26°C) to 7 days out (2.76°C).